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The Impact of Trade Credit Provision on Retail Inventory: An Empirical Investigation Using Synthetic Controls

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Abstract. Trade credit is an important source of short-term financing and an integrated part in supply contracts. Although a number of theories have been proposed on how trade credit could improve supply chain efficiency, causal studies on the impact of trade credit on operational decisions are scarce. In this study, we examine the impact of trade credit on inventory decisions using an empirical strategy that leverages (i) an exogenous shock imparted by the French government's intervention to impose a ceiling on trade credit duration, (ii) a triple difference-in-differences identification strategy, and (iii) synthetic controls (SCs). By considering the 60-day ceiling coverage and SC construction requirements, we identify four French retail sectors as our main sample. Among them, in the postregulation period, the hardware retail sector firms on average exhibited a significant 16% decline in trade credit usage. Correspondingly, these firms also displayed a significant 11% decline in inventory level. In the remaining three sectors, we found mixed results in the main sample. All the four sectors, however, show consistent support for a causal link between trade credit and inventory in a subsample compiled using a stringent 90-day ceiling criterion. Collectively, our findings offer direct evidence that trade credit is an indispensable financing source for inventory procurement. Finally, in the postregulation period, the hardware retailers exhibited a 15.5% decline in revenue and 3.5% reduction in gross profit. This cautions policy makers that regulations limiting the use of trade credit may have unintended consequences on downstream firms, and may harm overall supply chain efficiency.

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1. Introduction

Trade credit, which allows buyers in supply chains to delay paying their suppliers for an agreed period after receiving goods or services, is an important source of short-term financing (Petersen and Rajan 1997). A number of theories have been proposed to rationalize its prevalence (Chod et al. 2019b). Among them, several conjecture that trade credit has a unique advantage in achieving certain operational objectives, such as assuring product quality (Babich and Tang 2012), curtailing buyers' opportunistic operational decisions (Chod 2016), and sharing demand risk (Yang and Birge 2018). Collectively, these theories suggest that trade credit plays a crucial role in operational decisions.

An operational variable that has been closely linked to trade credit is inventory. As an effective lever to manage demand risk and improve supply chain performance, inventory is an important investment decision. According to Gaur et al. (2005), inventory accounts for 36% of total assets among U.S. public retailers. Given its practical importance, inventory management has long been at the core of operations management (OM). The theoretical investigation of the relationship between trade credit and inventory can be traced back at least to Haley and Higgins (1973), which shows that, under an economic order quantity (EOQ) model, trade credit terms and inventory decisions should be made jointly and that suboptimal payment terms could lead to suboptimal inventory stocking decisions. This view, combined with the aforementioned theories on trade credit, suggest that trade credit provision could have a material impact on firms' inventory decisions.

An alternative view, implied by the seminal work of Modigliani and Miller (1958), is that when the financial market is relatively efficient, trade credit, as a means of financing, does not necessarily have a significant impact on inventory, an operational decision. Under this view, trade credit does not play an essential role in financing inventory and can be readily substituted by other financing sources (e.g., bank credit).

The above two competing schools of thoughts create ambiguity with regard to the impact of trade credit provision on retail inventory levels. Given the importance of trade credit as a financial instrument and inventory as an operation decision, this paper aims to empirically investigate the causal impact of trade credit provision on retail inventory and its magnitude, which is absent in the literature.

To achieve this, we overcome two empirical challenges. The primary challenge arises from the embedded simultaneity between the trade credit and the inventory decisions (Haley and Higgins 1973). Presence of such simultaneity yields an endogeneity bias in estimates (Roberts and Whited 2013). We overcome this challenge by exploiting a policy intervention by the French government in 2008: Law No. 2008-776 on the Modernisation of the Economy (LME). This policy imparted an exogenous shock to trade credit provision by imposing a 60-day maximum repayment period for supplier-provided trade credit. As this law regulated only trade credit, and not inventory, any change in inventory associated with this regulation allows us to identify the causal relationship between trade credit and inventory. Furthermore, to control for the confounding effects of other shocks that overlap with the LME, for instance, the 2008 global financial crisis (GFC), we implement a triple difference-in-differences (DDD) identification strategy that exploits variation in the LME effect on the two decisions across sectors in France and the rest of the European nations.

The second challenge relates to the central requirement in studies using policies as exogenous shocks: constructing a credible counterfactual estimate for the outcomes of policy-affected entities in the absence of the policy intervention. The wide heterogeneity among European firms on various dimensions complicates the construction of a control group that can yield such a counterfactual estimate. In particular, in our context, the commonly used approaches (such as propensity score matching and coarsened exact matching) fail to eliminate the confounding impact of heterogeneity in group characteristics on the policy's impact on focal outcome variables. Specifically, under these approaches, we find strong empirical evidence toward violation of the central "parallel trends" assumption required for linear difference-in-differences (DD) estimation. We overcome the aforementioned challenge in constructing credible counterfactuals by using the relatively nascent synthetic control (SC) methodology (Abadie et al. 2010, Xu 2017), which has been described as "arguably the most important innovation in the policy evaluation literature in the last 15 years" (Athey and Imbens 2017). We adapt and extend the SC methodology to meet the demands of our large-size sample and DDD identification strategy. We validate our empirical strategy through two placebo analyses. One builds on selecting a pseudo treated nation among the control group nations, and the other selects a pseudo treatment year. Jointly, these placebo analyses ratify our DDD SC-based estimation strategy.

By considering the coverage of the LME and the number of control firms in other European countries that can be used for SC construction, we identified four French retail sectors, covering more than 5,000 firms, as our main sample. Across these four sectors, the retailers' pre-LME trade credit use varied considerably compared with the 60-day directive under the LME. As a result, in the post-LME period, the regulation constricted these retailers' trade credit usage to varying degrees, with some sectors showing a significant decline, and others continuing at the pre-LME levels. We find that as long as the LME leads to a significant drop in the retailers' trade credit usage in the post-LME period, there is consistent evidence of significant decline in these retailers' post-LME inventory levels. For example, in the hardware retail sector, we observe a significant decline of trade credit usage in the post-LME period. In this sector, we find that the LME reduced firms' average accounts payable by 14.1 days (approximately 16% of the pre-LME level). Correspondingly, firms also exhibited a significant reduction of 15.9 days of inventory, which is more than 11% of their inventory days before the LME. Together, these estimates suggest that a 1% reduction in trade credit resulted in a 0.67% decrease in inventory levels, establishing a sizable causal link between trade credit and inventory. In the other three retail sectors of our main sample, we observe mixed results (weak or no effect) regarding the impact of the LME on trade credit reduction.[1] Here, it is important to note that for the identification of the causal relationship between trade credit usage and inventory level, it is imperative that the LME regulation imparts a significant exogenous shock on the retailers' trade credit usage. Not surprisingly, we observe mixed results regarding the change on inventory level following the LME when considering the full sample of the three remaining sectors. However, when focusing on the subsample of retailers who are likely to be significantly affected in trade credit usage due to the LME (e.g., those with days of outstanding payables to be greater than 90 days in the pre-LME period), we find robust evidence of a causal link between trade credit and inventory similar to that in the hardware sector. We further validate this finding through a variety of robustness tests, including alternate subsamples, alternate variable definitions, alternate econometric models, and an alternate sample with an extended period.

Finally, we note that although the finding that restrictions on trade credit usage cause inventory declines is consistent with the extant theories on trade credit, there could exist alternative explanations to this phenomenon.

For example, facing pressure from trade credit reductions, firms could invest in improving operational efficiency and thus require less inventory without significantly sacrificing key financial performance. To shed light on the validity of these competing theories, we applied the same DDD SC method to two key financial metrics: retail revenue and gross profit. We find that in the hardware retail sector, the LME-caused trade credit and inventory reduction led to a 15.5% reduction in sales revenue and a 3.5% decline in gross profit. This finding provides additional support to theories on trade credit, which imply that restricting trade credit usage is likely to result in inventory underinvestment and poorer financial performance.

The contribution of this paper is twofold. First, to the best of our knowledge, this is the first empirical paper that establishes the causal relationship between trade credit provision and inventory, offering direct evidence that trade credit is an indispensable source of retail inventory financing. As such, this paper complements the existing theories on trade credit and, more generally, the OM–finance interface literature that emphasizes the importance of joint operational and financial decision making.

Second, our findings also bear important managerial and policy implications. By identifying the causal relationship between trade credit, inventory and the related financial metrics, our results suggest that when equipped with flexibility to design their trade credit–granting policies, suppliers need to take the associated operational and financial consequences into consideration. Similarly, our study also caution policy makers that regulations that limit the use of trade credit, such as the LME in France, with the intention to alleviate suppliers' financial burdens by accelerating payment,[2] could have a negative impact on the downstream party in the supply chain. As an efficient supply chain requires coordination of different parties, such unintended consequences could possibly hurt the overall supply chain efficiency.

2. Literature and Theoretical Background

Our paper is closely related to two streams of literature: (i) the theoretical literature on trade credit, on which our research question is grounded, and (ii) the empirical literature on trade credit and inventory, which our paper complements.

The theoretical literature on trade credit predominately studies the supplier's advantage in extending credit to their buyers, relative to financial institutions. See Peura et al. (2017) for a summary of such theories. Within this literature, our paper is most relevant to those that directly associate trade credit with inventory decisions. Based on the EOQ model, Haley and Higgins (1973) highlight the importance of coordinating trade credit terms and inventory decisions. They find that shorter trade credit terms lead to a less-than-optimal inventory level. Recently, Kouvelis and Zhao (2012) and Yang and Birge (2018) showed that, in a selling-to-the-newsvendor model, trade credit serves as an effective demand risk-sharing mechanism and induces the retailer to carry an inventory level closer to the system-optimal one. The direct implication of this theory is that if trade credit is made unavailable or restricted to a below-equilibrium level, the retailer will stock lower inventory, in turn hurting supply chain efficiency. This implication is also corroborated by some other theories of trade credit. For example, an implication in the work of Babich and Tang (2012) is that limiting trade credit may induce downstream buyers to carry less inventory due to concerns over product quality. Similarly, Biais and Gollier (1997) also imply that restricting trade credit usage weakens its information role and thus results in downstream firms' lower investment in inventory. Combined, these theories suggest that trade credit creates value in the supply chain and plays an indispensable role in inventory decisions that cannot be completely replaced by other financial means (e.g., bank credit) and/or operational levers (e.g., wholesale price).

An alternative view follows the line of reasoning in the seminal work of Modigliani and Miller (1958): when the financial market is relatively efficiency, the usage of trade credit, a specific source of financing, should have a negligible impact on operational decisions such as inventory. Given the two possibilities, the causal relationship between trade credit and inventory is ambiguous. Thus, the existence of this relationship and its magnitude are best answered empirically.

In the empirical literature, our paper is related to both studies on trade credit and inventory. Earlier empirical research on trade credit, such as that by Petersen and Rajan (1997), use survey or observational data to identify the determinants of trade credit terms, and they indirectly support the various roles of trade credit. In OM, Cai et al. (2014) find that bank and trade credit can be either substitute or complement in inventory financing. Lee et al. (2018) examine the correlation between trade credit, competition, and firm performance. In the finance literature, Barrot (2016) exploits French legislation that limits trade credit terms in the trucking sector to show that shorter terms lower barriers for new entrants. Similarly, exploiting SupplierPay—the U.S. government initiative that expedited government payments to their suppliers—Barrot and Nanda (2016) found that paying firms faster improves supplier balance sheets and their willingness to hire more employees. Both papers focus on the financial impact of trade credit usage and its subsequent effect on the economy, in service industries. Despite their significant role in the economy, these industries differ from our setting, as inventory considerations are absent when granting trade credit.

The paper that is most closely related to ours is that by Breza and Liberman (2017), and is, to the best of our knowledge, the only paper that empirically analyzes the impact of trade credit on the operational relationships between suppliers and buyers. Exploiting an agreement between the Chilean government and a large retail chain, the authors found that restricting trade credit led to a drop in the probability of trade, the price, and total sales between the large retailer and their small suppliers. Our work complements that by Breza and Liberman (2017) along two dimensions. First, our focus is on the downstream firm's inventory level, which is of particular importance to the OM community (Cachon and Terwiesch 2008). Second, our findings generalize beyond a single firm to a large number of retailers.

Our paper is also related to the empirical literature on inventory. Gaur et al. (2005), Rumyantsev and Netessine (2007), and Chen et al. (2007) offer comprehensive empirical analyses of inventory performance and its drivers in different industry sectors. Kesavan et al. (2010) focus on how inventory informs sales forecast. Jain et al. (2013) focus on inventory performance in a global sourcing setting. Hendricks and Singhal (2009) and Alan et al. (2014) show an association between inventory performance and equity returns. Wu et al. (2019) study the role of credit lines in inventory investment. Our paper contributes to this literature by identifying the causal relationship between trade credit and inventory.

Broadly speaking, our paper belongs to the literature on the interface of operations, finance, and risk management (Babich and Kouvelis 2018). The OM–finance interface literature has recently seen an increasing number of empirical works, including those by Osadchiy et al. (2015), Tunca and Zhu (2018), Serpa and Krishnan (2017), Wang et al. (2020), Xu et al. (2021), and Corsten et al. (2020). Methodologically, our paper complements this literature by introducing an adaptation of the SC methodology suited for large-size samples. Also, it emphasizes the importance of considering the operations–finance interaction in public policy making.

3. Empirical Setting: Identification and Estimation Strategy

In this section, we first describe the French government policy that imposes a ceiling on the trade credit period. Next, we provide details of our empirical strategy that leverages this policy intervention to identify and estimate the impact of trade credit on inventory decisions.

3.1. French Government Policy Intervention to Restrict Trade Credit

In August 2008, the French government enacted the Law on the Modernisation of the Economy with the aim of stimulating commerce by removing structural obstacles for small and medium enterprises (SMEs; Grall and Lamy 2009). Among other things,[3] the LME expanded an extant ceiling for payment terms of trade credit, introduced in 2006 for only the trucking industry (Barrot 2016), to all sectors. Specifically, the law restricted payment of trade credit to a maximum of 60 days (net) or 45 days (end-of-month) for transactions after January 1, 2009, with preagreed terms between a supplier and buyer. The law stipulates that French suppliers and buyers should comply with the mandatory payment terms, irrespective of their counterparty's location, foreign or domestic (Vincent 2009).[4]

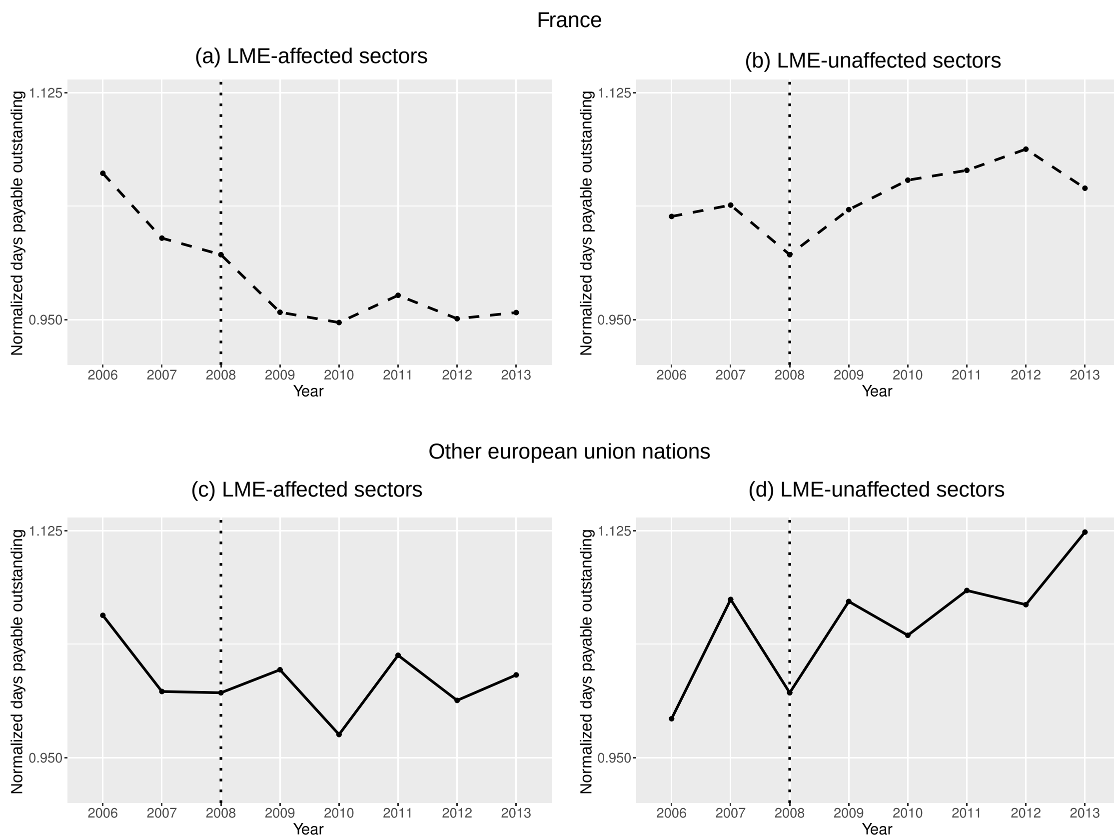
Naturally, not all sectors were equally constrained by the 60-day trade credit restriction imposed by the LME. Sectors where firms engaged in longer payment terms as a norm were most affected. For example, in the hardware retail sector, the average days payable outstanding (DPO; equal to accounts payable divided by costs of goods sold times 365) was 85.6 days before the LME.[5] Firms in some sectors were largely unaffected by the LME because their terms of trade were well below the 60-day ceiling. For example, firms in the fruit and vegetable retail sector, on average, repaid trade credit within 36.9 days even before the LME.

In Figure 1, we present model-free evidence of the impact of the French LME policy intervention on the trade credit received by the retailers.[6] We examine trends in both sectors that were affected by the LME (LME-affected sectors), defined as sectors with an average DPO, in 2008, for that sector's firms in France of at least 60 days, and sectors that most likely remained unaffected (LME-unaffected sectors), those with a sector average DPO in 2008 of less than 45 days. Panels (a) and (b) respectively capture the trend in retailers' DPOs. The vertical axis in each panel shows the average DPO as a unitless ratio, after normalizing each firm's DPO by its 2008 value. The horizontal axis covers an eight-year period spanning 2006 to 2013, which is split into the pre- and post-LME phases. The vertical dashed line highlights the end of the pre-LME phase with 2008 and the first financial year of the post-LME phase in 2009.[7] In panels (c) and (d), we present trends in trade credit usage in the rest of Europe by averaging respective trends in the aforementioned LME-affected and LME-unaffected sectors across six nations: Finland, Germany, Italy, Portugal, Spain, and Sweden.[8]

We find that French firms in the LME-affected sectors (panel (a)) experienced a persistent reduction in trade credit after introduction of the LME (i.e., between 2009 and 2013). In contrast, similar firms in other European nations (non-French nations) experienced a small increase in trade credit over this period (panel (c)). For the LME-unaffected firms, in both France and the non-French nations, we find, on average, an increase in DPO over time.

We next describe our identification strategy, which builds on the presence of the LME-affected and LME-unaffected sectors to identify the impact of trade credit on inventory decisions.

Figure 1. Trends in Payment Terms: Before and After LME



Notes. This figure shows trends in average DPO received by the retailers in the LME-affected and LME-unaffected sectors over an eight-year period between 2006 and 2013. Panels (a) and (b) present these trends for France. Panels (c) and (d) present these trends averaged across six European nations. The vertical axes show average DPO usage after normalizing each firm's DPO usage by its 2008 value.

3.2. Identification Strategy

We face a key challenge in identifying the impact of trade credit on inventory levels due to the embedded *simultaneity* in making trade credit and inventory decisions, which induces endogeneity bias (Roberts and Whited 2013). We exploit the exogenous shock imparted on trade credit provisioning decisions by the LME to overcome this simultaneity-in-decisions challenge. The law imposed an exogenous contraction in payment terms that, in turn, induced firms to make optimal adjustments to their inventory levels, which enables identification of the impact of trade credit on inventory levels.

In evaluating the impact of a policy intervention, a common concern is the presence of contemporaneous shocks that can affect focal variables of interest. For instance, in our context, the introduction of the LME overlaps with the aftermath of the 2008 GFC, a period with a severe credit crisis that also spilled over to firms' trade credit and inventory decisions (Alessandria et al. 2011). Henceforth, for the sake of brevity, we refer to all contemporaneous shocks as the GFC—the prominent known shock that overlaps with the LME. Next, we discuss our identification strategy to estimate the LME's impact while controlling for the GFC effect.

3.2.1. Identification Mechanism: DDD.

Our identification strategy rests on two observations. One, the LME constrained trade credit terms for France-based firms, but not for the other European firms.[9] Second, as highlighted above, there are sectors within France that were unaffected by the LME. Combining these observations, we implement a DDD strategy to isolate the effect of the LME on trade credit provisioning and resultant inventory decisions. To facilitate explanation, we illustrate our strategy using the following simplified setting.

Consider a two-period setting with comparable nations F and E. Nation F implemented the LME, which affected sector A but not sector U. We use the short notation $s \in \{FA, FU, EA, EU\}$ to denote affected and unaffected sectors across the two nations. Let the dependent variable y of firm i in sector s be defined by

$$\mathbb{E}[y_{ist} \mid s, t] = \beta_s + \beta_t + \beta_{s,t} + \beta_s^{\text{LME}} \times IS\_LME\_AFFECTED \times IS\_POST\_LME + \beta_s^{\text{GFC}} \times IS\_POST\_LME, \quad (1)$$

where β_s captures the sector-specific time-invariant component, β_t denotes the sector-invariant common temporal component, $\beta_{s,t}$ captures the sector-specific temporal component, β_s^{GFC} captures the sector-specific impact of the confounding GFC shock that overlaps with LME commencement, β_s^{LME} captures the LME policy's sector-specific impact, *IS_LME_AFFECTED* is a dummy variable that is set to one for the LME-affected sector A and zero otherwise, and *IS_POST_LME* is a dummy variable that is set to one for the post-LME period (abbreviated by *Post*) and zero for the pre-LME period (*Pre*).

Using Equation (1), we derive the double differences for sectors A and U ((A:DD) and (U:DD), respectively) and DDD estimates as

$$\{\mathbb{E}[y_{ist} \mid s = FA, t = Post] - \mathbb{E}[y_{ist} \mid s = FA, t = Pre]\}$$

$$- \{\mathbb{E}[y_{ist} \mid s = EA, t = Post] - \mathbb{E}[y_{ist} \mid s = EA, t = Pre]\} = \beta^{\mathrm{LME}}_{s=FA} + \beta^{\mathrm{GFC}}_{s=FA} - \beta^{\mathrm{GFC}}_{s=EA}, \quad \text{(A:DD)}$$

$$\{\mathbb{E}[y_{ist} \mid s = FU, t = Post] - \mathbb{E}[y_{ist} \mid s = FU, t = Pre]\}$$

$$- \{\mathbb{E}[y_{ist} \mid s = EU, t = Post] - \mathbb{E}[y_{ist} \mid s = EU, t = Pre]\} = \beta^{\mathrm{GFC}}_{s=FU} - \beta^{\mathrm{GFC}}_{s=EU}, \quad \text{(U:DD)}$$

$$[\{\mathbb{E}[y_{ist} \mid s = FA, t = Post] - \mathbb{E}[y_{ist} \mid s = FA, t = Pre]\} - \{\mathbb{E}[y_{ist} \mid s = EA, t = Post] - \mathbb{E}[y_{ist} \mid s = EA, t = Pre]\}]$$

$$- [\{\mathbb{E}[y_{ist} \mid s = FU, t = Post] - \mathbb{E}[y_{ist} \mid s = FU, t = Pre]\} \{\mathbb{E}[y_{ist} \mid s = EU, t = Post] - \mathbb{E}[y_{ist} \mid s = EU, t = Pre]\}]$$

$$= \beta^{\mathrm{LME}}_{s=FA} + [\beta^{\mathrm{GFC}}_{s=FA} - \beta^{\mathrm{GFC}}_{s=FU}] - [\beta^{\mathrm{GFC}}_{s=EA} - \beta^{\mathrm{GFC}}_{s=EU}]. \quad \text{(DDD)}$$

From Equation (A:DD), we see that if the GFC effect is constant across the two nations in the LME-affected sector (i.e., $\beta^{\mathrm{GFC}}_{s=FA} = \beta^{\mathrm{GFC}}_{s=EA}$), then the second difference estimate would suffice in identifying the impact of the LME on the focal variable y. In our setting, this assumption would imply that the GFC impact is constant across all European nations. In the absence of any concrete evidence, making such an assumption would be too restrictive.

Comparatively, in the case of the DDD estimate (Equation (DDD)), a relatively less restrictive assumption of a constant relative GFC impact between the LME-affected (A) and LME-unaffected (U) sectors across the two nations *enables* identification of the LME effect. After controlling for the nation-level differences of the GFC impact, the relative difference in its impact between sectors seemingly would reflect the differences in these sectors' business environments (e.g., the demand uncertainty of products being traded). Our placebo analyses in Section 7 provide support for the applied DDD strategy. Here, it is important to note that the DDD strategy retains the focal variable comparison (trade credit or inventory) between firms in alike sectors (i.e., between French and non-French firms in LME-affected and LME-unaffected sectors).

In summary, the combination of an exogenous shock due to the LME that resulted in trade credit tightening and the DDD framework enables us to identify the role of trade credit in inventory decisions while controlling for the aforementioned identification challenges.

3.3. Estimation Strategy: Synthetic Controls

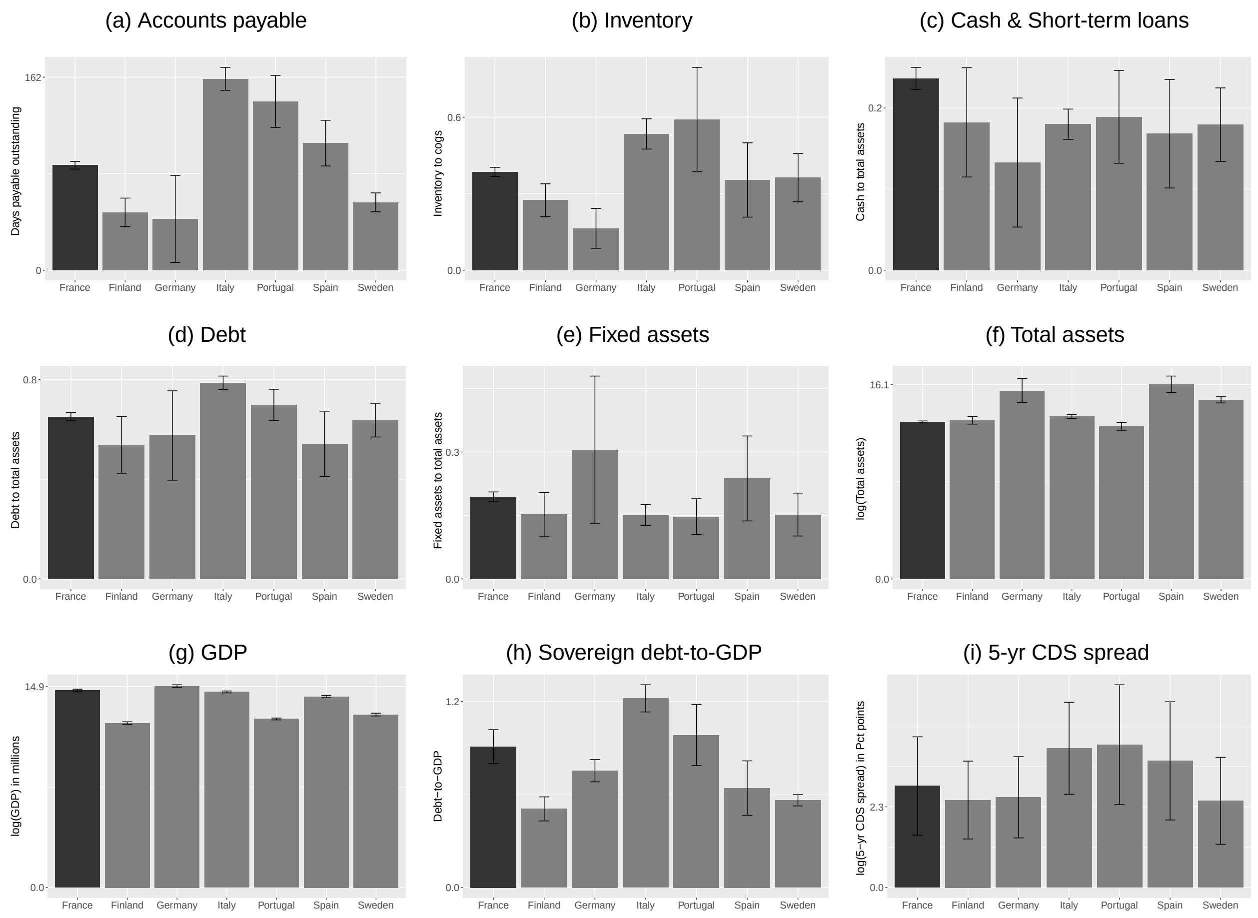
An important step in estimating the causal effect of a policy intervention using a DD framework is to construct a credible counterfactual path of policy-affected entities (treated firms) in the absence of policy intervention (Athey and Imbens 2017). In our context, the counterparts of LME-affected sectors in non-French nations provide a natural pool of control entities to construct the counterfactual path. However, among these non-French nations, there is no clear choice that best emulates French firm characteristics and their business environment overall. For instance, consider Figure 2, which presents comparisons of key firm-level (panels (a)–(f)) and macro-level (panels (g)–(i)) variables between France and select non-French nations for the hardware, paints, and glass retail sector (NACE code 47.52).[10]

As shown, firms in non-French nations exhibit contrasting patterns across variables in comparison with French firms. For instance, Italian firms exhibit much higher levels of accounts payable (panel (a)) compared with the French firms, but lower fixed assets (panel (e)), which typically proxies for a firm's investment in supply chain information technology capabilities (Gaur et al. 2005, Jain et al. 2013). In contrast, we observe the opposite characteristics from German firms: much lower payables and higher investment in fixed assets.

The aforementioned contrasting patterns of non-French nations suggest that, instead of an ad hoc selection of firms in one of the non-French nations as control entities, a more appropriate choice would be a blend of firms in these nations. For instance, hypothetical firms with blended characteristics of German and Italian firms may better mirror French firms. The SC estimation approach enables such a comparison by constructing an "optimally" blended entity based on prepolicy period characteristics (Abadie et al. 2010, Athey and Imbens 2017). Moreover, in contrast to the linear DD approach that requires a constant difference in the impact of the unobservable confounders (commonly referred to as the parallel trends assumption), the SC estimation permits for time-varying effects of such confounders.

Given the above-cited benefits of the SC methodology in our context, we employ it as our primary estimation method. In Section 5, we provide an overview of the SC estimation and its adaptation to our context. For robustness, in Online Appendix G, we provide results of a restrictive linear DDD estimation.

Figure 2. Firm and Country Characteristics



Notes. This figure shows bar charts of firm- and macro-level characteristics by country with their associated 99% confidence intervals. Firm characteristics (in panels (a)–(f)) are based on reported financials in 2006. Inventory is scaled by cost of goods sold. Cash and short-term loans, debt, and fixed assets are scaled by total assets. Total assets is reported in log form. Statistics of macroeconomics variables are computed using an eight-year period (2006–2013). GDP is reported as the log of GDP in millions, debt-to-GDP is the ratio of total federal debt to GDP, and the CDS spread is given as the log of the five-year sovereign CDS spread.

4. Data Description and Variable Construction

We construct the sample for this study by compiling data from three distinct sources: (i) European firm-level financials from the Bureau van Dijk (BvD) database, (ii) macroeconomic country-level data from the Organisation for Economic Co-operation and Development (OECD), and (iii) credit default spreads of European sovereign bonds from the Markit credit default swap (CDS) database.

4.1. Sample Data Construction

We extract European firm financials, public and private, from the BvD data, which are originally sourced from business registers maintained by local chambers of commerce. Kalemli-Ozcan et al. (2015) notes that the BvD data cover up to 75% to 80% of the economic activity reported in Eurostat,[11] making them a preferred data set for coverage of public and private European Union–based firms. The database provides information on a variety of balance sheet and income statement variables.

In this study, we focus on retail firms, which are identified by NACE codes 45.11, 45.19, 45.32, 45.40, and all codes within group 47. The BvD data cover a total of 66,758 firms across these retail sectors. The focus on retailers is for two reasons. First, as the firms in the last tier of a supply chain, these firms receive trade credit from upstream suppliers, but do not extend trade credit to their buyers (end consumers). This enables us to isolate the effect of trade credit provision from upstream firms and avoid any confounding factors due to firms' own considerations for extending trade credit to downstream partners. Second, unlike other sectors such as manufacturing, retail firms largely carry finished goods inventory. This allows us to focus on the impact of trade credit on the inventory directly associated with it.

Our study covers an eight-year period between 2006 and 2013. This choice of period was made for two reasons.

First, the BvD database's pre-2006 coverage is quite limited. For example, the number of retailers covered in 2005 is half that in 2006. Second, the European Union's Late Payment Directive mandated that its members bring their respective laws on trade credit management in line with Union regulations by 2013 (European Parliament and Council of the European Union 2011). To avoid any confounding effects from this European Union directive, we omit data after 2013.

We make a few choices toward the inclusion of sectors/firms in the final sample. These choices are for three reasons: (i) BvD data coverage characteristics, (ii) minimizing the impact of outlier firms (Barrot 2016), and (iii) improving the likelihood of constructing good-quality SCs.[12] First, we exclude firms that either have missing information or reported zero value for trade credit (accounts payable) and inventory levels before the LME. We observed that small firms exhibit a high propensity for missing data (over 55%), potentially on the account of inconsistencies in data reporting of these small firms or in coverage efforts of intermediaries through whom BvD periodically complies data. To adjust for such data coverage issues, we focus on firms with total assets of at least US$50,000, yielding 40,464 firms.

Second, we deal with outlier firms as in Barrot (2016). We remove firms that (a) report having trade credit periods of longer than a year and, analogously for our study, (b) report having more than one year's worth of inventory. The former condition indicates that the concerned firm is most likely under financial duress, which may also spill over to its other operational decisions. The latter condition focuses on excessively high inventory levels, which again suggests operational decisions being undertaken under unusual firm-specific idiosyncratic business conditions. Finally, to limit the impact of extreme values, while accounting for sector- and country-specific characteristics, we winsorize all our variables at the country-sector level using a 5% threshold (Verbeek 2008). The resultant sample has a total of 34,427 firms with an average DPO of 94.0 (respectively, 92.6) days among the French (respectively, non-French) firms.

Third, for enabling construction of well-fitted SCs, we focus on the LME-affected sectors that have a presence of at least 10 firms in at least five non-French nations. Note that the higher the SC's fit quality with the treated entities, the better the credibility of the counterfactual estimates. The SC's fit quality is naturally determined by the size and shape of the convex hull formed by the control entities. Thus, for firms in treated sectors that have a limited presence in non-French nations, the SC method may not be able to construct a well-fitted SC. We provide the count of non-French nations for each of the LME-affected sectors in Table A9 of Online Appendix F. Among all the LME-affected retail sectors, we find that the following four sectors meet the aforementioned criteria: (i) retail trade of motor vehicle parts and accessories (NACE code 45.32); (ii) retail sale of hardware, paints, and glass in specialized stores (code 47.52); (iii) retail sale of furniture, lighting equipment, and other household articles in specialized stores (code 47.59); and (iv) retail sale of clothing in specialized stores (code 47.71). We find that none of the LME-unaffected sectors by themselves satisfy the aforementioned criteria; thus, we pool firms in the following sectors to define our LME-unaffected sector: (i) retail sale of fruit and vegetables in specialized stores (code 47.21); (ii) retail sale of meat and meat products in specialized stores (code 47.22); (iii) retail sale of fish, crustaceans, and mollusks in specialized stores (code 47.23); and (iv) retail sale of automotive fuel in specialized stores (code 47.30).

The collective application of these inclusion choices yields the main analysis sample, with 7,103 firms in the four LME-affected sectors. Out of these, 5,494 firms are in France (treated firms), and 1,609 firms are in non-French nations (control firms). Similarly, for the LME-unaffected sectors, the sample includes a total of 2,517 firms, among which 2,125 are French and 392 are not French. We note that a higher representation of French firms in our sample is in line with extant studies (Dai 2012). This seems to be driven by both industry-structure differences and BvD's varying coverage across the European Union nations.[13] We cannot ascertain the extent of overrepresentation due to BvD's varying coverage. In line with this data limitation, in this study, we interpret the obtained treatment estimates as the average treatment effect on the treated entities rather than average treatment effect on the population.

4.2. Variable Construction: Dependent Variables

4.2.1. Trade Credit (DPO). Following the literature (Chod et al. 2019a), we use accounts payable (BvD variable *CRED*) to measure trade credit provided to retailers. Formally, for a firm k in period t, we measure the amount of trade credit received from its suppliers by DPO as

$$DPO_{kt} = \frac{CRED_{kt}}{MATE_{kt}} \times 365, \quad (2)$$

where *MATE* is the BvD variable for cost of goods sold (COGS).

4.2.2. Inventory Level (DSI). We measure a firm's inventory stocking decision using a relative inventory measure. Similar to Rumyantsev and Netessine (2007), we normalize inventory (BvD variable *STOK*) by COGS, deemed as a proxy of a firm's demand (Kesavan et al. 2010, Jain et al. 2013). Thus, one can interpret this relative inventory measure as a firm's optimal inventory stocking levels to expected demand. Formally, in line with the trade credit measure (*DPO*), we construct days sales of inventory

(DSI) as

$$DSI_{kt} = \frac{STOK_{kt}}{MATE_{kt}} \times 365. \quad (3)$$

4.3. Variable Construction: Covariates

For each of the two dependent variables, we include explanatory variables that are commonly studied in the literature to populate covariate vector $\boldsymbol{Z}$ of observables for SC construction (refer to Online Appendix B for details). The vector $\boldsymbol{Z}$ includes the following firm-level covariates: log of total assets (BvD variable *TOAS*), size-normalized measures of COGS, gross profit (equal to *OPRE* minus *MATE*), fixed assets (BvD variable *FIAS*), and long-term debt (equal to *TSHF* minus *SHFD*).[14] Note that variables on COGS, profit margins, and fixed assets have been studied in inventory-related studies respectively as proxies for demand, cost of underage, and capital investment in supply chain information technology (Gaur et al. 2005). These variables are also used in the trade credit empirical literature, along with variables on long-term debt and total assets. In addition, to capture the embedded simultaneity between the trade credit and inventory decisions, we include *DSI* and *DPO* variables, respectively, in the $\boldsymbol{Z}$ vector of DPO and DSI models for SC construction.

We complement the aforementioned firm-level covariates by augmenting the $\boldsymbol{Z}$ vector with the following country-level macroeconomic variables: log of GDP (OECD variable *GDP*), sovereign-debt-to-GDP ratio (general government debt), and log of the five-year CDS spread (Markit variable *spread5y*). Collectively, these variables provide a proxy for the overall strength of the financial and business environment (Ang and Longstaff 2013). In Table 1, we provide summary statistics of the above-defined firm-level variables.

5. Synthetic Control Estimation with Multiple Treatment Units

The central principle of the SC method is that a linear combination of control entities provides a better comparison for a treated entity than any single control. An SC unit is constructed for each treated entity by assigning weights to the available control entities. The weights are derived using prepolicy intervention data and assigned to minimize the difference between the treated entity and the associated SC with respect to both the outcome variable and observed covariates. In Online Appendix B, we provide an overview of the SC estimation methodology using a simple example of a single treatment unit. Below, we discuss its conceptual extension to multiple treatment units setting like ours. The implementation details on how we compute the DDD estimate in a large sample setting are provided in Online Appendix C.

5.1. Multiple Treatment Units

Two approaches are adopted in the literature to extend the single-treatment-unit SC methodology to the setting with multiple treatment units. The first approach is to aggregate both the treated and control entities at a higher level, which reduces the multiple treated entities to a single treated entity (Abadie et al. 2010). In our setting, implementing this approach would entail aggregation at the country level, which, in turn, would considerably reduce the analysis power because of a small country-level sample.

The second approach advocates for constructing an SC for each treated entity and using it to compute the treatment effect $\hat{\tau}$ for each of the treated entities. Specifically, we denote the individual treatment effect for firm k at period t as $\hat{\tau}_{kt}$, where $t = 1, \dots, T$, with the policy coming into force in period $t = T_o + 1 > 1$. Next, we compute the average treatment effect $\overline{\tau}$ by taking the weighted average of these individual treatment effects. The weights for each treated entity can be computed in multiple ways, including with (i) a simple equal-weighted allocation scheme (Xu 2017), (ii) entity-level characteristics (Kreif et al. 2016), and (iii) a goodness-of-fit criterion for constructed SCs (Acemoglu et al. 2016), the last of which we follow in this paper. Specifically, we compute the average treatment effect $\overline{\tau}$ of the LME as

$$\overline{\tau} = \frac{\sum_{k \in \text{Treatment Group}} \hat{\sigma}_k^{-1} \sum_{t=T_0+1}^{T} \hat{\tau}_{kt}}{\sum_{k \in \text{Treatment Group}} \hat{\sigma}_k^{-1}}, \quad (4)$$

where $\hat{\sigma}_k$ measures the goodness of fit using the root mean square of prediction error between the constructed SC and treated entity based on the preintervention period values of the outcome variable and covariates. Formally, $\hat{\sigma}_k$ is defined as

$$\hat{\sigma}_k = \sqrt{\frac{\sum_{t=0}^{T_0} \hat{\tau}_{kt}^2}{T}}. \quad (5)$$

Intuitively, the $1/\hat{\sigma}_k$ scaling factor provides larger weights to treatment effects that are estimated using better-fitted SCs. Following Acemoglu et al. (2016), we define a treated entity k as having a poor fit if $\hat{\sigma}_k \geq \hat{\sigma}\sqrt{3}$, where $\hat{\sigma} = \sum_{k=1}^{K} \hat{\sigma}_k / K$ is the average goodness of fit across all treated entities, and K is the number of firms in the treatment group. With this framework, we separately estimate the impact of the LME on trade credit and inventory levels for each of the four LME-affected sectors in our sample. This minimizes the effect of any sector-specific issues in using accounting variables to proxy for the actual contractual terms on trade credit repayment (Barrot 2016). Our unit of analysis is the firm-year level, $k \times t$. To cater to the unique requirements of our empirical study, we implemented two modifications to the classic SC estimation methodology: (i) a sampling approach to manage the large sample size computational complexity

Table 1. Summary Statistics: Treated vs. Synthetic Controls (Sector 47.52)

	French (956 firms)					Non-French (459 firms)				
	Mean	SD	25th	50th	75th	Mean	SD	25th	50th	75th
Trade credit (*DPO*)	85.6	37.8	55.8	82.7	111.3	119.8	68.8	59.4	115.3	166.6
Inventory (*DSI*)	142.9	79.4	80.3	135.8	191.7	171.6	148.8	73.1	129.3	220.7
Cash to assets	0.18	0.17	0.03	0.12	0.28	0.10	0.12	0.01	0.06	0.15
Fixed assets to assets	0.19	0.14	0.08	0.16	0.28	0.16	0.16	0.04	0.10	0.25
Debt to assets	0.63	0.2	0.49	0.65	0.79	0.70	0.22	0.56	0.76	0.88
Gross profit to assets	0.84	0.36	0.59	0.76	0.99	0.49	0.34	0.27	0.39	0.61
COGS to assets	1.20	0.45	0.85	1.16	1.50	1.17	0.64	0.73	1.05	1.46
log(Total assets)	13.04	1.05	12.27	13.03	13.83	13.82	1.47	12.75	13.57	14.79
	Synthetic controls									
	DPO					DSI				
	Mean	SD	25th	50th	75th	Mean	SD	25th	50th	75th
Trade credit (*DPO*)	88.2	20.2	76.42	86.3	99.0	—	—	—	—	—
Inventory (*DSI*)	—	—	—	—	—	150.8	52.2	117.4	145.8	180.8
Cash to assets	0.16	0.10	0.08	0.13	0.23	0.13	0.09	0.06	0.11	0.17
Fixed assets to assets	0.21	0.10	0.13	0.20	0.28	0.21	0.12	0.11	0.19	0.31
Debt to assets	0.61	0.14	0.51	0.61	0.73	0.65	0.12	0.56	0.67	0.85
Gross profit to assets	0.75	0.25	0.56	0.73	0.97	0.72	0.26	0.52	0.66	0.85
COGS to assets	1.33	0.46	1.01	1.28	1.50	1.32	0.43	0.98	1.32	1.63
log(Total assets)	13.78	0.78	13.20	13.73	14.22	13.85	0.76	13.27	13.87	14.40

and (ii) an extension to implement the DDD identification strategy. See Online Appendix C for the technical details.

6. Results

We estimate the LME's impact on the two focal outcome variables—trade credit (*DPO*) and inventory level (*DSI*)—by constructing separate SCs with each of them. Below, we present the findings using the analysis of sector 47.52 (retail sale of hardware, paints, and glass in specialized stores), which yields the best-fit SCs.[15] We provide the findings of the remaining three sectors in Section 6.3.

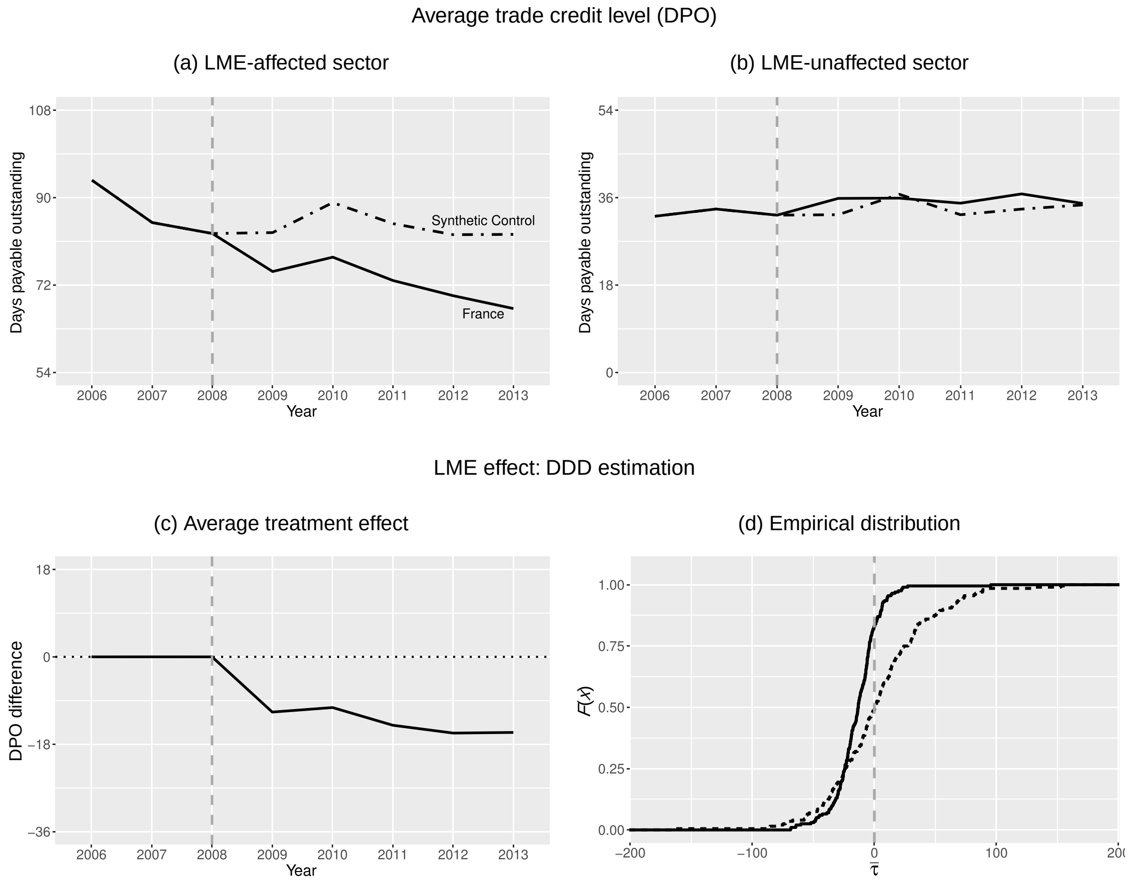
Table 1 presents summary statistics of key variables using the pre-LME-period observations. The median DPO among the French firms is 82.7 days, much above the 60-day LME restriction. Compared with the all-sectors relative trend in average trade credit usage across the French and non-French firms (94.0 versus 92.6 days), we find this sector exhibits an opposite trend of lower average trade credit usage among the French firms relative to their non-French counterparts (85.6 versus 119.8 days). In contrast to the pool of all control entities, constructed SCs exhibit characteristics that are more comparable to those of the treated entities. For instance, the absolute difference in the average DPO (respectively, DSI) is only 2.6 (7.9) days between the treated entities and the trade credit–based (inventory-based) SCs, whereas the difference is 34.2 (28.7) days between the treated group and the pool of all control entities. In SC construction, we find that, on average, higher weights are assigned to firms from Germany, Finland, Italy, and Sweden for both outcome variables (see panels (a) and (c) in Figure A9 in Online Appendix F). Collectively, these four non-French nations account for, on average, 0.8 and 0.85 of the total weight, respectively, in constructing trade credit–based and inventory-based SCs.

6.1. Trade Credit

Figure 3, illustrates the interim results under the SC methodology for sector 47.52. Panel (a) of Figure 3 shows the average DPO for the treated (solid line) and corresponding SC (dashed line) entities in the pre- and post-LME periods. Likewise, panel (b) presents average DPO for the French firms (solid line) and SC entities (dashed line) in LME-unaffacted sectors. Panel (c) presents a calibrated difference in the average DPO of treated and corresponding SC entities by differencing out any difference in the LME-unaffected sectors between France and non-French nations on account of the GFC (as discussed in the DDD extension in Online Appendix C). In other words, panel (c) presents the DDD estimate of the LME impact on trade credit provision. Finally, panel (d) shows the empirical cumulative distribution of estimated treatment and no-treatment effects using the DDD approach.

We find that, over the pre-LME period, the difference in average trade credit of treated and SC entities is close to zero. In the post-LME period, we find a persistent decrease in average trade credit of the treated entities. In

Figure 3. Synthetic Control Estimation: Impact of the LME on Trade Credit Provisioning in Sector 47.52



comparison, the average trade credit of the SCs shows a steady trend in the post-LME period, which is consistent with the model-free evidence of European firms in LME-affected sectors (panel (c) of Figure 1). The average change in the trade credit usage of hardware retailers based on the DDD procedure is –14.1 days. This corresponds to 16% decrease on the average pre-LME trade credit level of 85.6 days. Furthermore, we find this decline to be statistically significant. The 99% confidence interval (CI) based on the no-treatment effect distribution (see the discussion in Online Appendix C) is [–4.7, 10.7]. Furthermore, the Kolmogorov–Smirnov (K–S) test rejects the null hypothesis of the treatment and no-treatment effect distributions being identical with a p-value less than 0.001.

6.2. Inventory Level

In Figure 4, similar to the analysis on trade credit, we provide an illustration of the SC estimation results. We find that, compared with the preintervention period, all firms exhibited an increase in average inventory levels during the post-LME period (panels (a) and (b)). This indicates that common macroeconomic factors over time resulted in a steady temporal increase in inventory levels. However, similar to the patterns observed in the model-free graphs (Figure A1 in Online Appendix A), the treated firms in France show a much smaller increase in inventory level compared with their counterpart SCs. In contrast, in the LME-unaffected sector, the increasing inventory trends are much more comparable between the French firms and the corresponding SCs. Together, these observed asymmetries in the DSI usage trend suggest a spillover impact of LME on inventory decisions due to significant decline in trade credit usage of French hardware sector retailers. Specifically, panel (c) shows a significant impact of the LME in reducing average inventory levels of the treated entities. The DDD estimate is –15.9 days (99% no-treatment effect CI, [–7.0, 22.5]). This reflects a significant decrease of 11% on the average pre-LME inventory levels of 142.9 days. Because the LME does not directly influence inventory, this decrease is attributed to the reduction of trade credit received by the retailers on account of the LME. The significance of these estimates is also corroborated by the K–S test rejection of the null hypothesis that the treatment and no-treatment effect distributions are identical, with a p-value less than 1%.

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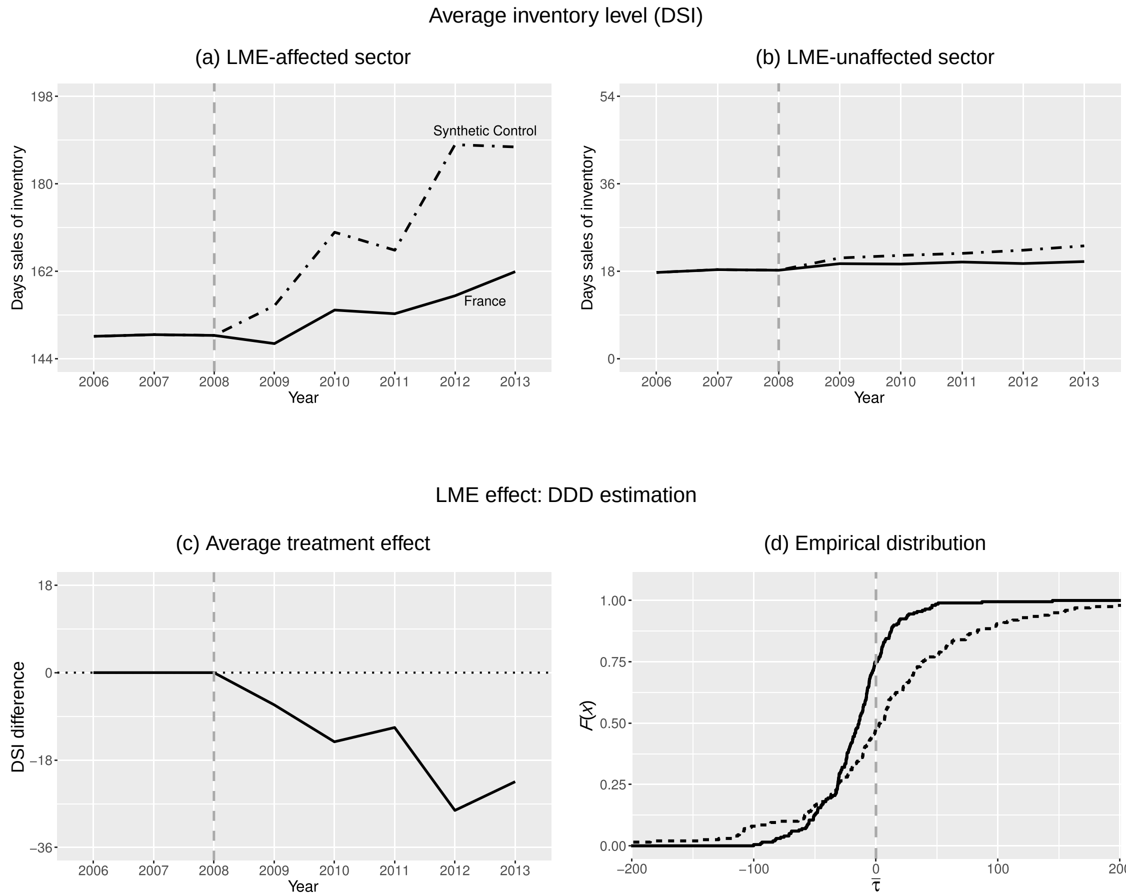
Figure 4. Synthetic Control Estimation: Impact of the LME on Inventory Levels in Sector 47.52



In summary, the above analysis on the impact of the LME on trade credit and inventory levels provides strong empirical evidence for the causal role of trade credit in determining inventory level in the French hardware retail sector. Put differently, the exogenous shock by the LME enforcement resulted in these retailers experiencing a significant reduction in the amount of trade credit they receive from their suppliers. In response, these firms lowered their inventory level drastically. Combining the estimates of trade credit decline and inventory drop, our results suggest that a 1% decrease in trade credit leads to a 0.67% decline in inventory.

6.3. Results of the Three Remaining Sectors

So far, we have focused on presenting the findings based on the hardware retail sector. We also replicated the same DDD analysis on the other three remaining sectors that satisfy the basic inclusion criterion for the main sample (see Section 4.1), namely, 45.32, 47.59, and 47.71. The detailed results of these three sectors are presented in Online Appendix D. In short, for sector 47.71 (specialized clothing retailers), we find that the LME leads to a significant drop in trade credit (–6.2 days, with a 99% CI of [3.7, 22.2]) and inventory levels (–17.4 days, with a 90% CI of [–17.3, –1.7]). For sectors 45.32 (motor vehicle parts retailers) and 47.59 (future and specialized household stores), we do not observe a significant decline in trade credit. The apparent absence of significant LME impact on the retailers' trade credit usage in these sectors is likely due to a mixture of two factors. First, compared with 47.52, we find that in these sectors, a larger fraction of retailers were already complaint with the 60-day LME directive. In other words, for many retailers in these sectors, the pre-LME trade credit usage was already below the 60-day threshold. Second, we find that the constructed SC counterfactuals of the retailers in these sectors are of a poorer quality. Motivated by these observations, for these sectors, we replicated our analysis with subsamples of retailers with longer DPO in the pre-LME period (e.g., >60 days) and found support for significant DPO and DSI declines in the post-LME period (see Table A6 in Online Appendix D). In summary, we find that across the retailers in our main sample, as long as the LME leads to a significant drop of trade credit usage due to LME restriction, there is consistent evidence that the decline of trade credit is associated to a decrease of inventory.

6.4. Synthetic Controls: Strengths and Limitations

Our analysis reveals a balanced view on the strengths and limitations of the SC methodology, which is still in its nascent development stages. On the one hand, the SC methodology can provide a viable alternative to examine quasi-natural experiments. In such settings, researchers prefer to apply the classical linear difference-in-differences estimation but are often constrained to finding controls that satisfy the required parallel trends assumption. The SC methodology relaxes this requirement. On the other hand, the efficacy of SC methodology, as illustrated in our analysis, relies on the ability to construct good-quality SCs, which, in turn, depends on available observables and the length of pretreatment period data. Yet other challenging aspect in using the SC methodology could be the sample size of treated and control units. As discussed in Appendix C, the current packages may take hours to find optimal weights for SCs depending on control group size. Our proposed sampling approach could be effective in circumventing the sample size challenge. Finally, the SC methodology can also be limiting in obtaining heterogeneous treatment effects, as it is onerous to replicate moderating variable analysis within the current SC methodology framework.

7. Robustness

In this section, we present robustness tests for our main findings using alternate criteria for sample creation, alternate variable definitions for outcome variables, alternate criteria for aggregation of multiple treatment effects, and alternate econometric models. In addition, we present results with alternate control groups to test the sensitivity of observed finding, (i) on the premise that the stable unit treatment value assumption (SUTVA; Abadie and Cattaneo 2018) holds true in our context, and (ii) on the chosen vector of covariates. We further examine whether our findings that are based on the hardware retail sector can be generalized to the other retail sectors. In addition to the result based on the three sectors summarized in Section 6.3, we replicate our analysis on an extended sample of six retail sectors obtained by modifying our main sample construction criteria.[16] Across these multiple robustness tests, we continue to find strong support for our findings, both within the hardware retail sector and across a range of other retail sectors. We conclude this section by describing the results of two placebo tests that support the effectiveness of the proposed DDD framework in identifying the LME-caused effects on trade credit and inventory levels.

7.1. Alternate Sample Construction

We first test whether our findings are sensitive to the sample construction choices that were made to curtail the impact of data coverage and outlier issues (for details, see Section 4). Specifically, in rows 2 and 3 of Table 2, we present estimates using samples constructed respectively with alternate minimum thresholds of US$100,000 and US$25,000 on firm size, measures that proxy for coverage extent and missing values in the BvD data set. In row 4, we show results with a sample that reflects changes in the winsorization level from 5% to 1%—a relaxed definition of extreme value. We reproduce estimates using the main sample in row 1 for comparison.

Next, to enable construction of better fit quality SCs, our main sample included the LME-affected sectors that have at least 10 firms in at least five non-French nations. We test robustness of the obtained finding to this criterion by reducing the minimum number of non-French nations to four. This expands our sample to 10 sectors, including the four sectors of our main sample. Row 5 reports results using a pooled sample of the six additional sectors: (i) other retail sale in nonspecialized stores (NACE code 47.19), (ii) retail sale of electrical household appliances in specialized stores (code 47.54), (iii) retail sale of newspapers and stationery in specialized stores

Table 2. Robustness Analyses

No.	Means test DPO	Means test DSI	K–S test DPO	K–S test DSI	# affected (Trt, Con)	# unaffected (Trt, Con)	Robustness test description
1	−14.1***	−15.9***	−14.1***	−15.9***	(956, 459)	(2,125, 392)	Main sample estimates
2	−15.7**	−13.9***	−15.7**	−13.9***	(860, 435)	(1,533, 341)	Min. total assets ≥ $100k
3	−13.8**	−15.8***	−13.8***	−15.8***	(992, 462)	(2,587, 455)	Min. total assets ≥ $25k
4	−9.9***	−19.0*	−9.9***	−19.0**	(956, 459)	(2,125, 392)	Winsorization at 1%
5	−3.9***	−13.4***	−3.9***	−13.4***	(5,571, 1,051)	(2,125, 392)	Six sector pooled sample analysis
6	−16.3***	−30.8***	−16.3***	−30.8***	(2,734, 114)	(1,169, 31)	Extended period pooled: 2005–2013
7	−11.7***	−12.4***	−11.7***	−12.4**	(947, 399)	(2,123, 390)	Normalization by sales
8	−13.1***	−11.3***	−13.1***	−11.3***	(956, 459)	(2,125, 392)	SC equal weighted
9	−14.2***	−16.1***	−14.2***	−16.1***	(956, 459)	(2,125, 392)	SC poorly fitted threshold $> \sigma\sqrt{2}$
10	−10.9***	−10.4***	−10.9***	−10.4***	(956, 1,150)	(2,125, 392)	SUTVA: alternate control group

Notes. This table displays the DDD SC estimates for the impact of the LME on trade credit provisioning (DPO) and inventory levels (DSI) across robustness tests. Statistical inference is based on the no-treatment distribution, with both the test of means and distribution shown. Trt, Treatment; Con, control.

*$p < 0.1$; **$p < 0.05$; ***$p < 0.01$.

(code 47.62), (iv) retail sale of sporting equipment in specialized stores (code 47.64), (v) retail sale of watches and jewelery in specialized stores (code 47.77), and (vi) other retail sale of new goods in specialized stores (code 47.78).

Finally, as noted before, the BvD data set coverage decreases sharply as we increase the pre-LME years in our study period. For example, our main sample covers the 2006 to 2013 period with 7,103 firms. In comparison, the periods starting with 2005 and 2004 respectively cover 4,084 and 827 firms. Intuitively, the SC fit quality is dependent on the number of pretreatment years. Thus, we examine sensitivity of our finding using the extended period of 2005 to 2013 period. Row 6 reports results of this test.

7.2. Alternate Variable Definition and Aggregation Criterion

In the main analysis, following the OM literature, we analyze relative inventory with respect to demand as measured using COGS (Gaur et al. 2005). Likewise, we measure trade credit provisioning by normalizing accounts payable using COGS. In row 7 of Table 2, we present robustness to this normalization choice by using sales, commonly used in trade credit literature (Barrot and Nanda 2016), as the normalizing factor. Next, we examine the sensitivity of estimates to the weighted-average method for collating multiple treatment effects to estimate an average treatment effect. In the main analysis, we determined weights using a goodness-of-fit scaling factor (Acemoglu et al. 2016). In row 8 of Table 2, we present estimates using an equal-weighted approach (Xu 2017). Row 9 shows the robustness of our findings toward the definition of poorly fitted SCs by changing the minimum quality threshold to $\sigma\sqrt{2}$ from $\sigma\sqrt{3}$.

7.3. Alternate Control Groups

Our main analysis builds on the premise of the SUTVA holding true in our context. It entails that trade credit usage by the non-French firms in the LME-affected sector—the control entities in the main analysis—is not impacted by the LME. However, it is conceivable that these firms are indirectly affected; for example, French suppliers may demand similar short repayment terms or exploit benefits of early payment by French buyers to gain a competitive edge in non-French markets by extending longer trade credit.[17] Thus, we test the robustness of our findings to the choice of non-French firms as control entities. Specifically, we construct SCs for the firms in sector 47.52 by using the non-French firms in the remaining three LME-affected sectors (45.32, 47.59, and 47.71) as control entities. In row 10 of Table 2, we present the estimates using this alternate SCs construction approach.

Next, we test the robustness of our findings to the choice of observables included in the covariate vector $\boldsymbol{Z}$. Though we draw on the extant empirical literature focusing on inventory and trade credit decisions to select these variables, data constraints limit us from including all factors that influence inventory decisions in the covariate vector $\boldsymbol{Z}$ (e.g., the demand uncertainty factor studied by Rumyantsev and Netessine (2007)). In Table A11 in Online Appendix F, we present estimates obtained with SCs constructed by excluding one covariate at a time. In all 16 tests, we find that results are consistent—both in direction and significance—with our main findings.

7.4. Alternate Econometric Models

Our main analysis focuses on the linear change in the focal outcome variables and uses the SC methodology to construct the counterfactual of treated entities. In the past, OM scholars have also used econometric models that capture nonlinear changes in inventory decisions (Gaur et al. 2005). Below, we present analysis using proportional growth–based outcome variables to test the robustness of the observed findings to this modeling choice. Furthermore, in Online Appendix G, we present results with an alternate approach to estimating the counterfactual. Specifically, we obtain ordinary least squares estimates under the classic linear DDD approach (Gallino et al. 2016, Staats et al. 2016, Dhanorkar 2019).

Similar to the main analysis, we implement a DDD strategy in three steps to identify the LME impact when modeling nonlinear change in the outcome variables. First, we construct a measure, of trade credit ($\widehat{DPO}$) and inventory ($\widehat{DSI}$) that capture the LME-induced effect after controlling for the firm and macro-level covariates $\boldsymbol{Z}$ that influence a firm's DPO and DSI decisions (see Section 4.3).[18] Second, following Tsoutsoura (2015), we measure the proportional growth in firm i's outcome variable y as the ratio of the average $\widehat{y}$ in the pre- and post-LME periods. Formally, $py_i = \frac{1}{5}\sum_{t=2013}^{2009}\widehat{y}_{it} / \frac{1}{3}\sum_{t=2006}^{2008}\widehat{y}_{it}$, where $y \in \{DPO, DSI\}$, and py denotes the proportional growth in outcome variable y. Effectively, the measure py captures the first (nonlinear) difference between the pre- and post-LME periods in variable $\widehat{y}$. Finally, we estimate the LME impact on the outcome variable by estimating the following linear model: $py_i = \alpha_0 + \alpha_A \mathbb{I}_{\text{LMA}} + \alpha_U \mathbb{I}_{\text{LMU}} + \alpha_{\text{LME}} \mathbb{I}_{\text{LMA}} \times \mathbb{I}_{\text{LMU}}$, where $\mathbb{I}_{\text{LMA}}$ (respectively, $\mathbb{I}_{\text{LMU}}$) denotes a dummy variable that is set to one for the LME-affected (respectively, LME-unaffected) sector, and α_{LME} captures the DDD estimate of the LME impact on $\widehat{y}$. We find α_{LME} equals -0.166 and -0.183, respectively, for the outcome variables *pDPO* and *pDSI*, both significant at 1% level. This implies that our results are robust irrespective of the modeling choice in outcome variables.

7.5. DDD Identification Strategy: Placebo Analyses

We perform two placebo analyses to validate the applied DDD framework. In particular, we examine whether GFC distorts identification of the LME-caused effects.

The first test examines whether a *placebo-treated* nation (i.e., a non-French nation that enforced a pseudo-LME in 2008) exhibits significant treatment effects under our applied empirical strategy. We use the remaining non-French nations as *placebo-control* nations, which collectively constitute the pool of control entities for SC construction. Note that because both the placebo-treated and placebo-control nations are affected by the common GFC shock, one would expect estimates of the pseudo-LME to be zero.

We perform the aforementioned placebo analysis on the following six non-French nations: Finland, Germany, Italy, Portugal, Spain, and Sweden. In 9 out of 12 estimates (6 for trade credit and 6 for inventory; Table A10 in Online Appendix F), we find no significant impact of the psuedo-LME passage on trade credit and inventory decisions at 10% level. The exceptional cases occur when either the outcome variables or the explanatory variables for the selected pseudo-treated nation are at the boundary of the convex hull of available control entities. For example, as shown in Figure 2, Italian firms exhibit the highest average trade credit levels, Portuguese firms have the highest average inventory levels, and Finnish firms have the lowest average for the Debt, GDP and sovereign-debt-to-GDP predictors. These boundary characteristics naturally limit construction of well-fitted SCs. Intuitively, for a given treated entity, constructing a comparable weighted-average SC is not feasible if the heterogeneity among the available pool of control entities does not encompass the treated entity's characteristics. Not surprisingly, in these cases, we find that the quality of constructed SCs is considerably poorer and, hence, yields significant estimates for the impact of the pseudo-LME.

The second test examines whether the French LME-affected firms exhibit significant treatment effects with a *placebo-LME* year in the pre-LME period.[19] Specifically, we replicate our analysis using the 2005–2008 period observations and by setting 2007 as the placebo-LME year. We find insignificant DDD estimates for DPO (–9.0; 90% no-treatment CI, [–34.8, 34.4]) and DSI (10.9; 90% no-treatment CI, [–52.4, 48.4]).

8. Financial Implications of Inventory Reduction

Thus far, we have established that trade credit provision has a significant impact on retail inventory. Although such evidence is consistent with extant trade credit theories predicting that limiting trade credit below the equilibrium level should result in inventory underinvestment, it may also occur through a different economic channel. For example, facing financing pressure due to a lack of trade credit, the conventional inventory financing source, retailers may invest in inventory efficiency and consequently require less inventory while maintaining similar levels of financial performance.

To shed light on such alternative explanations, we turn to two key financial performance metrics: revenue and gross profit. In general, trade credit theories predict that inventory underinvestment caused by trade credit restriction will cause revenue to drop significantly (Yang and Birge 2018). Similarly, such theories also predict that when trade credit is restricted to below the equilibrium level, especially when the supplier market is relatively competitive, the retailer's operational profit should also be negatively affected (Burkart and Ellingsen 2004, Chod 2016). On the other hand, if the reduction in inventory is caused by increased inventory efficiency, it should not lead to a simultaneous drop in both revenue and gross profit.

We put these competing hypotheses to empirical tests by using our main DDD synthetic control–based estimation methodology to investigate the impact of the LME-led trade credit and inventory reductions on revenue and gross profit. We normalize these two outcome variables by each firm's total assets. We find that after LME, the average change in the size-normalized revenue among French hardware retailers is –0.32 (99% no-treatment CI, [–0.07, 0.17]).[20] We note that in the pre-LME period, the average size-normalized revenue for French hardware retailers is 2.04. This implies that the LME resulted in a 15.5% decline in revenue.

Similarly, we find that the change in size-normalized gross profit is –0.032 (99% no-treatment CI, [–0.02, 0.03]), which translate to a 3.5% decline relative to the average pre-LME gross profit level (the pre-LME size-normalized gross profit is 0.85).[21] Collectively, the above results are consistent with existing trade credit theories, which imply that restricting trade credit could hurt financial performance. Furthermore, by comparing the changes in revenue and gross profit, we note that the inventory reduction caused by trade credit restriction affects revenue more severely than gross profit. One possible explanation is that the retailer may use other means (such as renegotiating the wholesale price, or increasing the retail price) to mitigate the impact of limited inventory on profitability. However, such mitigating measures are likely to affect upstream firm profitability, consumer welfare, or both.

These results also bear important managerial and policy implications. On the managerial side, they suggest that suppliers, when possessing the flexibility to determine trade credit terms, need to take the operational as well as the associated financial consequences into consideration. For policy makers, over the last decade or so, trade credit has become the target of several recent political and regulatory reforms around the world. In addition to the French LME, the U.S. government launched two initiatives in recent years, QuickPay and SupplierPay, which aim to accelerate trade credit repayments from the government and companies to their contractors and suppliers (Mandelbaum 2011, White House 2014). The European Union and United Kingdom have imposed similar regulations (European Parliament and Council of the European Union 2011; UK Department for Business, Energy and Industrial Strategy 2022). Mainly aiming to ease the financial burden of extending trade credit on SMEs, these reforms are well intended. However, our findings suggest that policies that restrict trade credit to a below-equilibrium level could have unintended negative consequences through operational channels.

9. Conclusion

Trade credit is an important external financing source for businesses. Related theoretical work proposes that trade credit affects not only the financial situations of firms offering and receiving it, but also operational decisions. In this paper, by using a French policy that restricts the trade credit period available to buyers, we offer a formal causal analysis of the impact of trade credit on inventory, a decision of vast importance in operations management. We find that among French hardware retailers, whose trade credit and inventory levels are significantly influenced by the LME, a 1% drop in trade credit leads to a 0.67% decline in inventory level. Furthermore, we find a significant decrease in revenue and profitability of these retailers, suggesting that restriction in trade credit usage and the concomitant retail inventory reduction can have considerable economic impact. Combined, our results provide strong evidence for the existing theories that emphasizes trade credit as an indispensable source in inventory financing and an important lever for supply chain efficiency. Furthermore, they alert managers and policy makers that a holistic approach is needed when designing trade credit–related policies.

Our study can be extended along several dimensions. First, because of data limitations, the scope of our study is limited to aggregate-level evidence among French retailers. Should more detailed data become available, further research that examines the exact mechanism (e.g., demand risk sharing or deterring opportunistic behaviors) through which trade credit leads to decline in inventory and profitability could better shed light on the validity of different theories on trade credit. On a related note, our quantification of trade credit impact on inventory is based on the French hardware retail sector. Studies using data from other countries and/or industries could help validate the generality of our findings. Second, this study focuses on the impact of the trade credit restriction on downstream inventory. Examining other operational metrics, such as service quality or upstream inventory, could offer a more comprehensive view of the impact of trade credit on operational decisions. Third, although we use the LME as the policy intervention, the goal of this study is not to provide a formal policy evaluation. However, our empirical findings suggests that a comprehensive policy evaluation should include both the benefits of trade credit for retailers and its potential harm on suppliers' operations.

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Endnotes

[1] The absence of a significant LME impact on the retailer's trade credit usage in these three sectors is likely due to a mixture of low trade credit levels before the LME and poorer SC quality. See Section 6.3 for a more detailed discussion and Online Appendix D for the technical details.

[2] Similar regulations in the past decade or so include two initiatives in the United States, QuickPay and SupplierPay (Mandelbaum 2011, White House 2014); the European Union Late Payment Directive (Directive 2011/7/EU; European Parliament and Council of the European Union 2011), and the United Kingdom's Prompt Payment Code (UK Department for Business, Energy and Industrial Strategy 2022).

[3] The LME also introduced guidelines on areas unrelated to trade credit, such as improving the standing of Paris as a financial center, simplifying the business environment for entrepreneurs, and incentivizing innovation.

[4] Marolleau (2016) notes that administrative penalties for violation of the LME terms can be levied "when the entire business relationship takes place in France," even if the contract between two firms is governed by the law of a foreign state.

[5] The LME imposed varying ceilings on the following five sectors to account for their strong seasonality imprint: (i) agricultural equipment (ceiling of 110 days end-of-month (EOM)); (ii) sporting goods—snow sports (90 days); (iii) leather goods (54 days EOM); (iv) watches, jewelery, and goldsmithing (74 days); and (v) toys (75–95 days depending on time of year). See Decree No. 2015-1484 of November 17, 2015 (Decree is from Le ministre de l'économie, de l'industrie et du numérique).

[6] Figure A1 in Online Appendix A presents equivalent model-free evidence of the LME impact on inventory levels.

[7] We account for the in-year enactment of the LME by mapping financial reporting to the calendar year as follows: for firms with a fiscal year ending in the period between July of year X to June of year $X + 1$, the financial numbers are mapped to year X.

[8] As detailed in Section 6, we find these six countries to be more comparable to French firms than the other European nations.

[9] We note that the European Union introduced a similar trade credit–tightening directive to its members that had to be integrated into their respective national laws by March 16, 2013, at the latest (Directive 2011/7/EU, the Late Payment Directive; European Parliament and Council of the European Union 2011). The actual implementation, however, stretched up to 2014 (e.g., Germany adopted the directive in July 2014).

[10] NACE (Nomenclature of Economic Activities revision 2) codes are created and maintained by Eurostat to classify economic activities of businesses across Europe.

[11] Eurostat provides economic activity statistics based on national censuses but does not provide the underlying firm-level data themselves.

[12] In Section 7, we show that none of our core findings are sensitive to the choices made in sample construction.

[13] Kalemli-Ozcan et al. (2015) reports that BvD covers 85.2% of total firms in French economy, 63.6% of firms in Germany, and 41.7% of firms in Spain.

[14] BvD variables *OPRE*, *TSHF*, and *SHFD* measure, respectively, operating revenue, total shareholder funds and liabilities, and total shareholder funds.

[15] As evident from Figure A4 in Online Appendix D, sector 47.52 stochastically dominates the three remaining sectors in the fit quality of the constructed SCs. In addition, as detailed later, sector 47.52 experiences a significant drop in trade credit usage following the LME, a prerequisite for our identification. The other three sectors, on the other hand, exhibit mixed evidence on this measure, weakening the basis of our identification.

[16] Another potential sector-specific concern is that maybe the hardware retail sector experienced a unique spillover impact of the GFC, which is closely related to the housing sector. If true, it would cast doubt on whether the support for our finding in the 47.52 sector is on account of best fit SCs or a confounding GFC effect. In Online Appendix E, we show evidence that mitigates concern of such a unique spillover impact.

[17] However, we note that, according to Vincent (2009), under the LME, both the French suppliers and buyers are expected to comply with the mandatory payment terms, irrespective of their business partner's location—foreign or domestic.

[18] Note that $\widehat{y_{it}} = y_{it} - \overline{y}_{it}$, where $\overline{y}_{it}$ is the estimate of y_{it} obtained by estimating the following linear model: $y_{it} = \beta_0 + \beta_z Z + \beta_i \mathbb{I}_i + \beta_t \mathbb{I}_t + \epsilon_{it}$, where β_i and β_t denote firm and time fixed effects.

[19] We thank the review team for suggesting this robustness check.

[20] We find these results to be robust across different subsamples. Specifically, for retail sectors where the LME causes a significant drop in trade credit, revenue and gross profit also experience sizeable declines. Table A13 in Online Appendix F reports results of these robustness tests.

[21] We note that the SC fit quality for revenue and gross profit is poorer by an order of 20 or more compared with the analyses with DPO and DSI. Unfortunately, the limited BvD coverage of financial variables, such as short-term debt, hinders the ability to improve SC fit quality by including well-documented predictors of revenue and profitability in the extant literature. As a result, our estimates of the financial outcome variables are more prone to the disproportional impact of outliers (treated entities with poor SC fit). For example, if we exclude outliers, from both sides, at the 5% level (respectively, 10%), the average revenue reduction equals 10.7% (respectively, 8.3%), and profit reduction equals 5.3% (respectively, 3.6%).

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